



Reference (4)

Jeff (CHI-HSUAN HO)

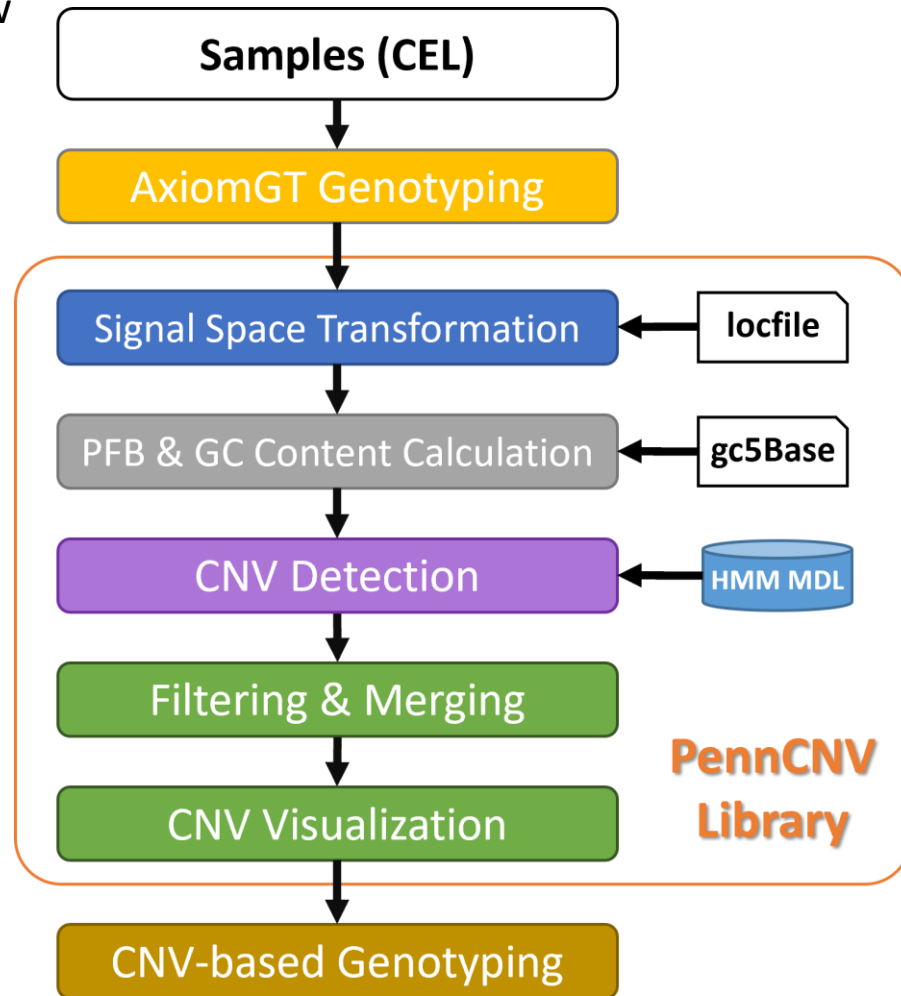
The subsequent slides only showcase a few parts of the work materials and results.



Bioinformatics Analyses – CPT Copy Number Variation (CNV) Analysis and PennCNV

Jeff (CHI-HSUAN HO)

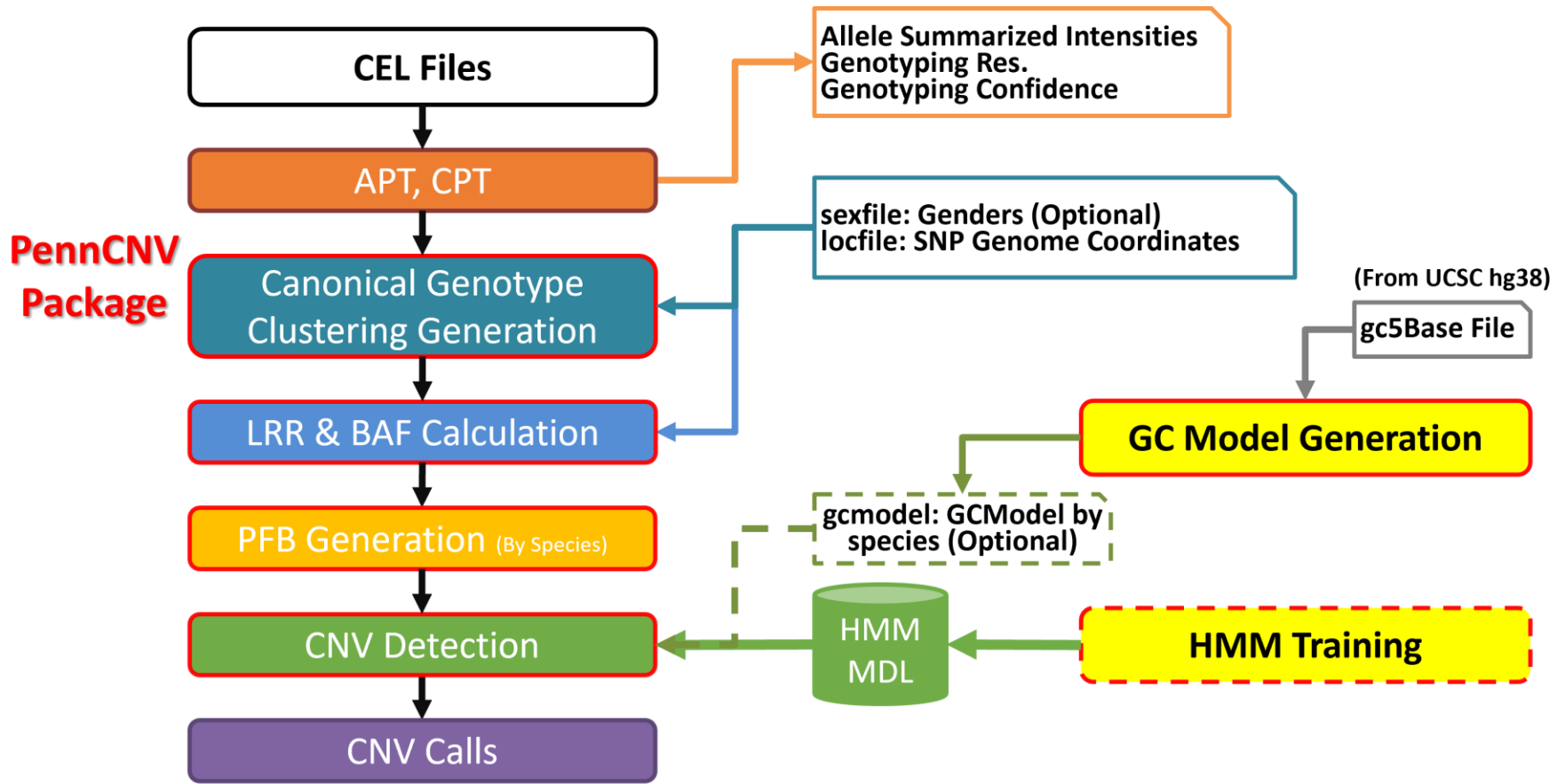
- CNV Analysis Workflow



CNV Analysis and PennCNV



- PennCNV Analysis Workflow and I/O



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CNV Analysis and PennCNV



- PennCNV Analysis Workflow and I/O

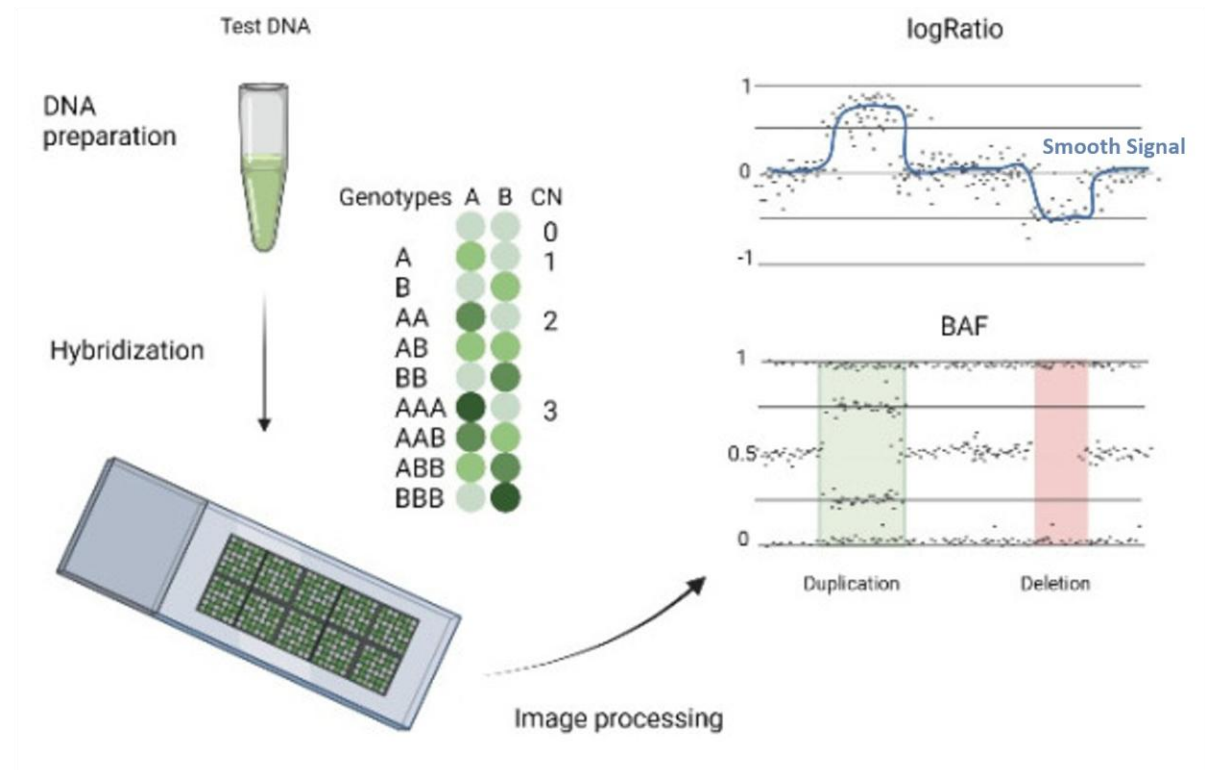
LRR & BAF Calculation

CNV Calls

Infer SNP Allele

Filter CNV

Downstream Programs



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CNV Analysis and PennCNV



- PennCNV Analysis Workflow and I/O

CNV Calls

Table 1. Hidden states, copy numbers, and their descriptions

Copy no. state	Total copy no.	Description (for autosome)	CNV genotypes
1	0	Deletion of two copies	Null
2	1	Deletion of one copy	A, B
3	2	Normal state	AA, AB, BB
4	2	Copy-neutral with LOH	AA, BB
5	3	Single copy duplication	AAA, AAB, ABB, BBB
6	4	Double copy duplication	AAAA, AAAB, AABB, ABBB, BBBB

Each state has a different distribution of CNV genotypes.

SNP Genome Coordinates From PFB file

of SNPs are contained within the CNV

Length of the CNV (Coordinates end – start + 1)

state: HMM State (Internal HMM Parameter),
cn: Actual Copy Number (Normal Autosomal: 2, Del.: 0 or 1, Dup.: >= 3),
Original File name,
startsnp: Starting marker identifier (SNP),
endsnp: Ending marker identifier (SNP)

chr3:37940970-37944758	numsnp=3	length=3,789	state2	cn=1	father.txt	startsnp=rs9837352	endsnp=rs9844203
chr3:75379524-75519068	numsnp=7	length=139,545	state2	cn=1	father.txt	startsnp=rs4677005	endsnp=rs2004089
chr11:81792950-81806219	numsnp=9	length=13,270	state2	cn=1	father.txt	startsnp=rs7947005	endsnp=rs12293984
chr20:10511631-10583260	numsnp=10	length=71,630	state2	cn=1	father.txt	startsnp=rs8114269	endsnp=rs682562
chr11:55603545-55679951	numsnp=11	length=76,407	state2	cn=1	mother.txt	startsnp=rs2456022	endsnp=rs7934845
chr11:549119-558884	numsnp=4	length=9,766	state5	cn=3	mother.txt	startsnp=rs4963136	endsnp=rs2061586
chr11:55603545-55669650	numsnp=8	length=66,106	state1	cn=0	offspring.txt	startsnp=rs2456022	endsnp=rs17498926
chr3:3957986-4054960	numsnp=50	length=96,975	state2	cn=1	offspring.txt	startsnp=rs11716390	endsnp=rs17039742
chr11:81792950-81806219	numsnp=9	length=13,270	state2	cn=1	offspring.txt	startsnp=rs7947005	endsnp=rs12293984
chr20:10511631-10583260	numsnp=10	length=71,630	state2	cn=1	offspring.txt	startsnp=rs8114269	endsnp=rs682562

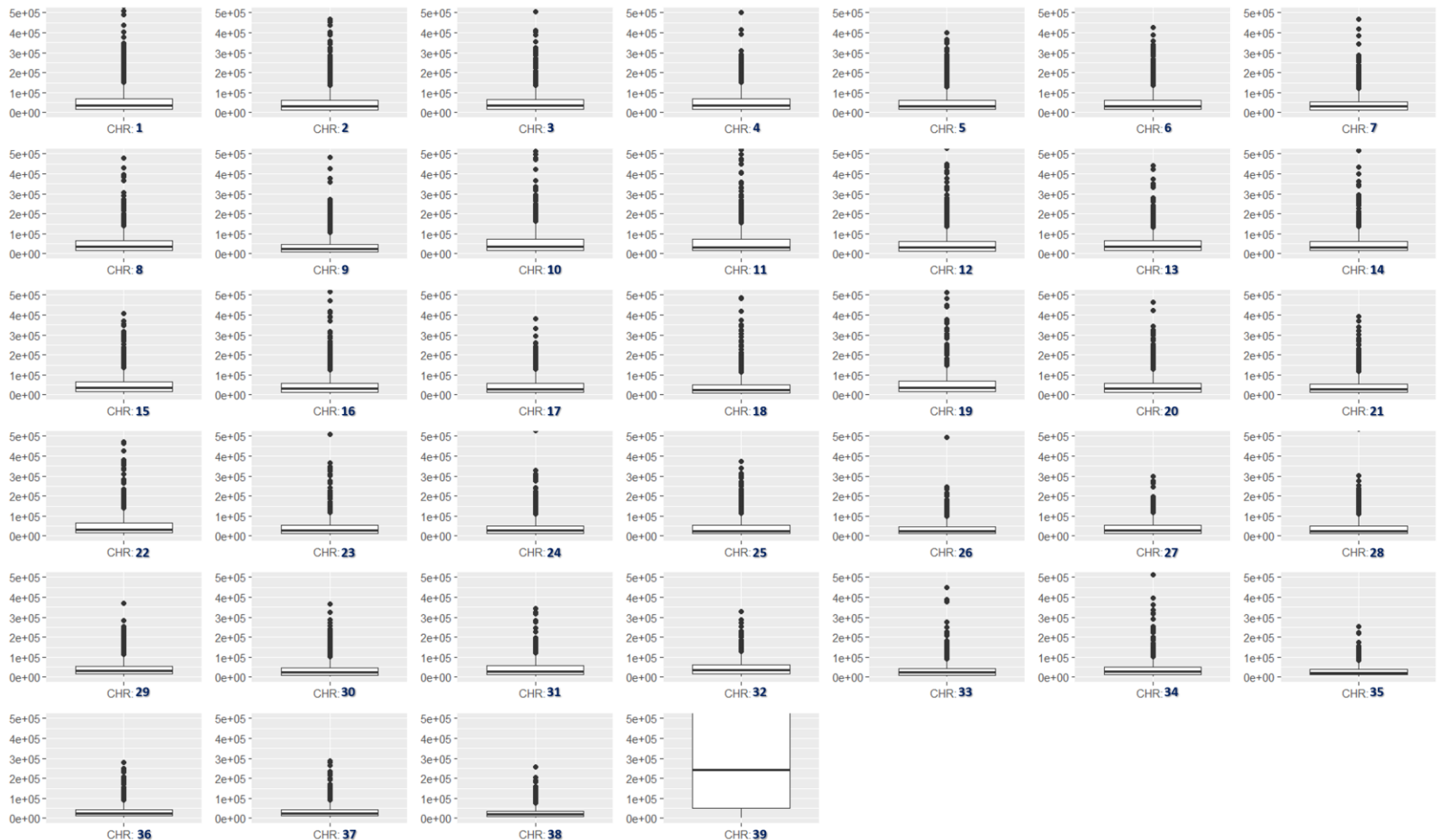
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CNV Analysis and PennCNV



- Genomic Distance Dist. Between Adjacent SNP Markers for Real Data (Banff Dog vcf)



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- Hidden Markov Model

- Emission Probability of log R ratio

$$P(r|Z) = \pi_r + (1 - \pi_r)\phi(r; \mu_{r,z}, s_{r,z})$$

- Emission Probability of B allele frequency

$$\begin{aligned} P(b|Z) = & \pi_b + (1 - \pi_b) \sum_{g=2}^{K(Z)-1} BN[g-1; K(Z)-1, p_B] \phi(b; \mu_{b,g}, s_{b,g}) \\ & + (1 - \pi_b) BN[0; K(Z)-1, p_B] [I_{\{b=0\}} M_0 \\ & + I_{\{0 < b < 1\}} \phi(b; \mu_{b,1}, s_{b,1})] + (1 - \pi_b) BN[K(Z)-1; K(Z)-1, p_B] \\ & [I_{\{b=1\}} M_1 + I_{\{0 < b < 1\}} \phi(b; \mu_{b,K(Z)}, s_{b,K(Z)})] \end{aligned} \quad (5)$$

where

$$BN[g-1; K(Z)-1, p_B] = \binom{K(Z)-1}{g-1} p_B^{g-1} (1 - p_B)^{K(Z)-g}$$

- Transition Probabilities of Hidden States

$$P(Z_i = l | Z_{i-1} = j) = \begin{cases} 1 - \sum_{k=2}^6 p_{j,k-1} (1 - e^{-d_i/D}), & \text{if } l = j \\ p_{j.,l-1} (1 - e^{-d_i/D}), & \text{if } l \neq j \end{cases} \quad (6)$$

CNV Analysis and PennCNV



- Real Data Performance Visualization and Evaluation

Reasonable

Unreasonable

Unknown,
Filtered by
PennCNV
default
params

Column	Sample_Name	Chr	Start_Pos	End_Pos	Cc	Max_Lo	Length_bp	No_P	LRR_mean	LRR	LRR_SD	BAF_mean	BAF_median	BAF_SD	BAF_drift	WF	NumCNV	Quality_Score
873	BanfiDog.165_20230118123031	19	46625504	49653801	3	139.369	3028297	76	-0.0002	0	0.1573	0.4981	0.5	0.0748	0.00741	0.0256	11	1
1444	BanfiDog.171_20230118121824	9	48181850	50667443	3	75.122	2485593	111	-0.0053	0	0.2062	0.497	0.5	0.0702	0.004897	-0.0483	24	1
3029	BanfiDog.261_20230118125321	14	56687500	57500155	3	50.09	812655	20	-0.0032	0	0.1495	0.4978	0.5	0.0641	0.003995	-0.0241	5	0.999999999
62	BanfiDog.103_20230116120933	21	45185418	46521857	3	51.972	1336439	36	-0.0109	0	0.2044	0.4975	0.5	0.0712	0.0088	-0.0315	11	0.999999999
849	BanfiDog.163_20230118123519	2	80689206	81639842	3	51.311	950636	46	-0.0025	0	0.2143	0.4969	0.5	0.0721	0.008907	-0.0527	66	0.999999994
483	BanfiDog.137_20230116122953	31	37244574	39046841	3	54.903	1802267	72	0.0013	0	0.2185	0.4964	0.5	0.0759	0.010337	-0.0442	37	0.999999985
4356	BanfiDog.77_20230118114005	34	27961770	31631537	1	116.975	3669767	84	-0.0004	0	0.181	0.4987	0.5	0.0694	0.00339	0.0301	8	0.999999947
4381	BanfiDog.99_20230118114212	9	48836667	49242478	3	40.282	405811	25	-0.0035	0	0.1816	0.4968	0.5	0.0655	0.004991	-0.0387	31	0.999999773
3028	BanfiDog.261_20230118125321	14	55644280	56170442	3	38.458	526162	17	-0.0032	0	0.1495	0.4978	0.5	0.0641	0.003995	-0.0241	5	0.999999712
574	BanfiDog.137_20230118120240	2	79699224	81618258	3	53.821	1919034	87	-0.0029	0	0.2784	0.4939	0.5	0.1061	0.023796	0.0416	150	0.999999971
4384	BanfiDog.99_20230118114212	28	39663235	40968873	3	42.943	1305638	52	-0.0035	0	0.1816	0.4968	0.5	0.0655	0.004991	-0.0387	31	0.9999999231
346	BanfiDog.109_20230118115401	34	27961770	31631537	1	108.821	3669767	84	-0.0002	0	0.1704	0.4986	0.5	0.0655	0.00342	0.0289	4	0.999999873
706	BanfiDog.141_20230118120703	24	46306925	47535413	3	43.385	1228488	48	0.0035	0	0.1998	0.4959	0.5	0.0844	0.014535	0.0294	53	0.999998925
4063	BanfiDog.69_20230116115056	6	46238543	46724138	3	34.565	485595	10	0.0085	0	0.1682	0.4983	0.5	0.0654	0.005244	0.0253	7	0.9999998474
350	BanfiDog.171_20230118121824	28	39667379	41164216	3	41.378	1496837	55	-0.0053	0	0.2062	0.497	0.5	0.0702	0.004897	-0.0483	24	0.999997963
79	BanfiDog.103_20230118115920	24	45990414	47535413	3	42.374	1544999	58	-0.0014	0	0.182	0.4969	0.5	0.0666	0.007274	-0.0385	27	0.999997933
4413	BanfiDog.99_20230118115415	35	6114985	6858763	3	35.327	743778	14	0.0067	0	0.1561	0.4966	0.5	0.0757	0.010898	-0.0214	24	0.999997612
24	BanfiDog.101_20230118114431	24	46055938	47535413	3	41.057	1479475	56	-0.0037	0	0.2017	0.4974	0.5	0.071	0.0064	-0.0435	33	0.999996909
44	BanfiDog.101_20230118115642	8	7196072	72562428	3	34.392	60356	21	0.0012	0	0.1489	0.498	0.5	0.0693	0.008528	-0.0247	21	0.999994694
3420	BanfiDog.269_20230116134739	24	46538449	47007004	3	36.279	468555	25	0.0111	0	0.1921	0.496	0.5	0.0792	0.014116	-0.0294	69	0.999994567
449	BanfiDog.133_20230118121512	24	46455041	46993637	3	34.81	538596	27	0.004	0	0.1517	0.4981	0.5	0.069	0.008208	-0.025	23	0.999991329
3894	BanfiDog.301_20230116140429	4	45861443	47285707	3	43.612	1424264	57	0.0184	0	0.2238	0.4923	0.5	0.1014	0.026978	0.0379	200	0.999990603
3938	BanfiDog.301_20230116140429	9	48345772	49242478	3	41.008	896706	44	0.0184	0	0.2238	0.4923	0.5	0.1014	0.026978	0.0379	200	0.999990419
495	BanfiDog.137_20230116122953	9	50004859	51289049	3	36.667	1284190	44	0.0013	0	0.2185	0.4964	0.5	0.0759	0.010337	-0.0442	37	0.999987825
3835	BanfiDog.299_20230118131328	24	46379308	47469083	3	38.601	1089775	45	0.0047	0	0.2688	0.4929	0.5	0.1026	0.02177	0.0436	136	0.999985079
4377	BanfiDog.99_20230118114212	24	45925798	47535413	3	38.761	1609615	61	-0.0035	0	0.1816	0.4968	0.5	0.0655	0.004991	-0.0387	31	0.999984052
3360	BanfiDog.267_20230116134447	31	37244574	38921288	3	39.491	1676714	65	0.0001	0	0.2256	0.4964	0.5	0.0721	0.009767	-0.0426	47	0.999981874
265	BanfiDog.109_20230116121716	9	48305033	49242478	3	39.14	937445	47	0.0193	0	0.2223	0.4951	0.5	0.0868	0.020708	0.036	122	0.99998155
4048	BanfiDog.301_20230116140429	26	28667272	29579246	3	36.42	911974	27	0.0184	0	0.2238	0.4923	0.5	0.1014	0.026978	0.0379	200	0.999980308
3670	BanfiDog.299_20230116140110	2	84271695	84812502	3	34.597	540807	31	0.0083	0	0.1935	0.4954	0.5	0.0837	0.015524	-0.0316	85	0.999973866
4233	BanfiDog.75_20230116115920	28	40097392	41164216	3	35.727	1066824	41	-0.0018	0	0.2488	0.4942	0.5	0.0926	0.026322	-0.0379	171	0.999970834
96	BanfiDog.103_20230118115920	9	48666617	49141968	3	31.031	475351	28	-0.0014	0	0.182	0.4969	0.5	0.0666	0.007274	-0.0385	27	0.999958155
3888	BanfiDog.301_20230116140429	20	56549816	58077727	3	34.658	1527911	34	0.0184	0	0.2238	0.4923	0.5	0.1014	0.026978	0.0379	200	0.999911553
4394	BanfiDog.99_20230118114212	2	84239419	84569593	3	27.819	330174	21	-0.0035	0	0.1816	0.4968	0.5	0.0655	0.004991	-0.0387	31	0.99990235
4338	BanfiDog.99_20230118114845	6	45357595	46061413	3	25.297	703818	11	-0.0017	0	0.1449	0.498	0.5	0.0673	0.004992	0.0042	8	0.999826784
794	BanfiDog.163_20230118123519	20	54693250	56038738	3	28.277	1345488	30	-0.0025	0	0.2143	0.4969	0.5	0.0721	0.008907	-0.0527	66	0.999824646
1454	BanfiDog.171_20230118121824	20	55791363	58077727	3	31.555	2286364	51	-0.0053	0	0.2062	0.497	0.5	0.0702	0.004897	-0.0483	24	0.999814135
3417	BanfiDog.269_20230116134739	18	45433107	45545179	3	27.489	112072	15	0.0111	0	0.1921	0.496	0.5	0.0792	0.014116	-0.0294	69	0.999801422
1352	BanfiDog.167_20230118121353	2	84271695	84656758	3	28.703	385063	24	-0.0012	0	0.2034	0.4975	0.5	0.0902	0.015274	-0.0308	41	0.999796022
88	BanfiDog.103_20230118115920	8	7196072	72562428	3	26.481	626356	21	-0.0014	0	0.182	0.4969	0.5	0.0666	0.007274	-0.0385	27	0.999791803
2833	BanfiDog.235_20230118124631	2	84271695	84569593	3	26.936	297898	19	-0.0006	0	0.171	0.4973	0.5	0.0752	0.009142	-0.0274	24	0.999779892
53	BanfiDog.101_20230118115642	20	55480506	55833410	3	24.838	352904	10	0.0012	0	0.1489	0.498	0.5	0.0693	0.008528	-0.0247	21	0.999744567
2814	BanfiDog.233_20230118124421	37	29796237	29967515	3	24.16	171278	9	-0.0033	0	0.1666	0.4966	0.5	0.0702	0.006184	-0.0271	12	0.999742455
3655	BanfiDog.299_20230116140110	19	26031466	27512287	3	28.779	1480821	26	0.0083	0	0.1935	0.4954	0.5	0.0837	0.015524	-0.0316	85	0.999738484
828	BanfiDog.163_20230118123519	24	46328677	47469083	4	30.444	1144046	46	-0.0025	0	0.2143	0.4969	0.5	0.0721	0.008907	-0.0527	66	0.999689002
1455	BanfiDog.171_20230118121824	8	72166982	72865792	3	24.576	698810	21	-0.0053	0	0.2062	0.497	0.5	0.0702	0.004897	-0.0483	24	0.999653507
229	BanfiDog.109_20230116121716	38	22017822	22318571	3	28.77	300749	25	0.0193	0	0.2223	0.4951	0.5	0.0868	0.020708	0.036	122	0.99952989
4184	BanfiDog.75_20230116115920	37	26872751	27830514	3	30.592	957763	42	-0.0018	0	0.2488	0.4942	0.5	0.0926	0.026322	-0.0379	171	0.999526535
16	BanfiDog.101_20230118114431	28	40047432	41164216	3	29.051	1116784	44	-0.0037	0	0.2017	0.4974	0.5	0.071	0.0064	-0.0435	33	0.999512838
3578	BanfiDog.291_20230118130423	6	45474836	46724138	3	24.064	1249302	20	-0.0045	0	0.2196	0.4964	0.5	0.0716	0.009205	0.0343	37	0.999503417
133	BanfiDog.105_20230118114842	2	84271695	84711957	3	27.755	440262	27	-0.0025	0	0.2292	0.4938	0.5	0.095	0.017242	0.0352	83	0.999497771
2774	BanfiDog.227_20230118123730	28	40337780	40697573	3	24.343	359793	17	-0.0008	0	0.1456	0.4977	0.5	0.0642	0.004692	-0.0264	10	0.999485658

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CNV Analysis and PennCNV

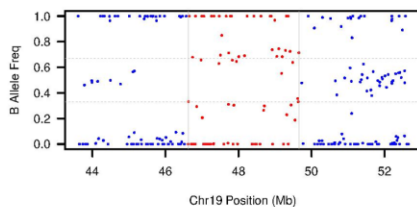
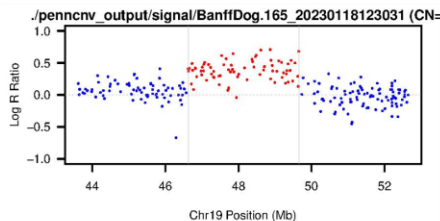


- Real Data Performance Visualization and Evaluation

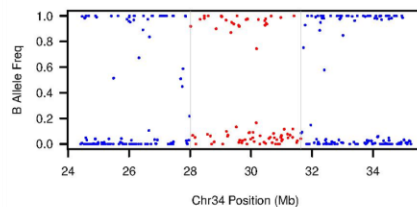
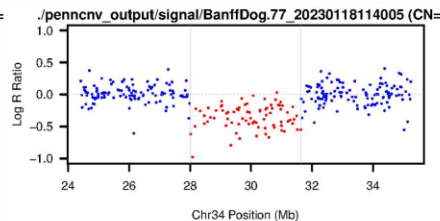
- CNV Analysis for Banffdog Data.
 - 7 possible CNV calls / total 111 calls

ID	Sample_Name	C	Start_Pos	End_Pos	Cd	Max_L	Length	No	LRR_r	LR	LRR	BAF	BA	BAF_d	WF	N	Quality_Sc	
56	BanffDog.165_20230118123031	19	46625504	49653801	3	135.886	3028297	74	0	0	0.1585	0.4979	0.5	0.074	0.007177	0.026	3	1
100	BanffDog.77_20230118114005	34	28020211	31631537	1	117.59	3611326	80	4.00E-04	0	0.1722	0.4988	0.5	0.067	0.003023	0.028	1	0.980651002
28	BanffDog.109_20230118115401	34	27961770	31631537	1	106.753	3669767	82	-0.0011	0	0.1621	0.4988	0.5	0.063	0.002653	0.028	1	0.907155866
87	BanffDog.261_20230118125321	14	56687500	57500155	3	59.955	812655	20	-0.0025	0	0.1385	0.498	0.5	0.062	0.003485	-0.022	3	0.999808186
17	BanffDog.103_20230116120933	21	45185418	46521857	3	53.253	1336439	36	-0.0105	0	0.2035	0.4982	0.5	0.07	0.009312	-0.031	1	0.999038921
88	BanffDog.261_20230118125321	14	55523883	56170442	3	49.801	646559	18	-0.0025	0	0.1385	0.498	0.5	0.062	0.003485	-0.022	3	0.998318725
97	BanffDog.69_20230116115056	6	46238543	46724138	3	36.632	485595	10	0.0088	0	0.1625	0.4984	0.5	0.065	0.005049	0.025	1	0.943835832

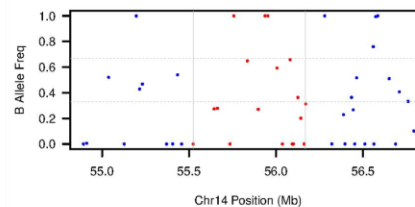
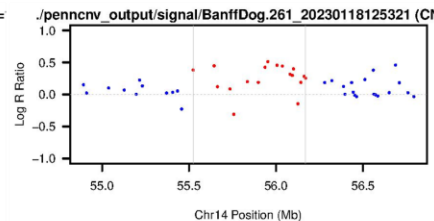
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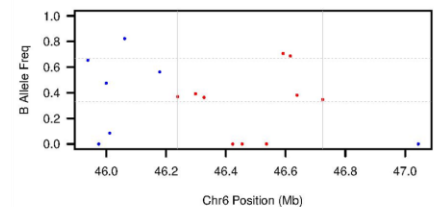
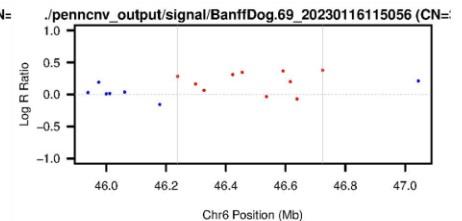
CN=3, conf = 135.8



CN=1, conf = 117.59



CN=3, conf = 49.8



CN=3, conf = 36.6

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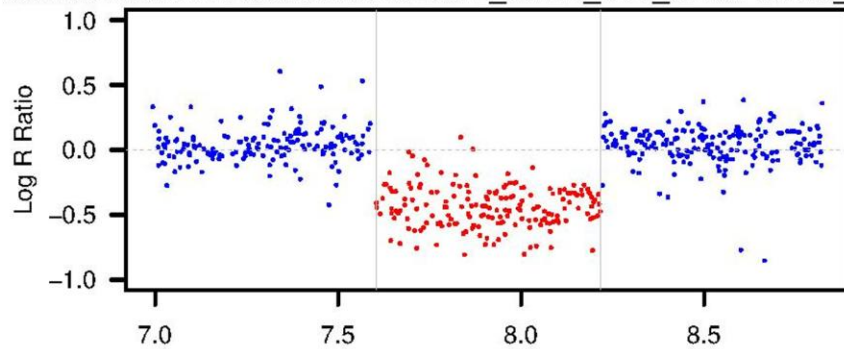
CNV Analysis and PennCNV



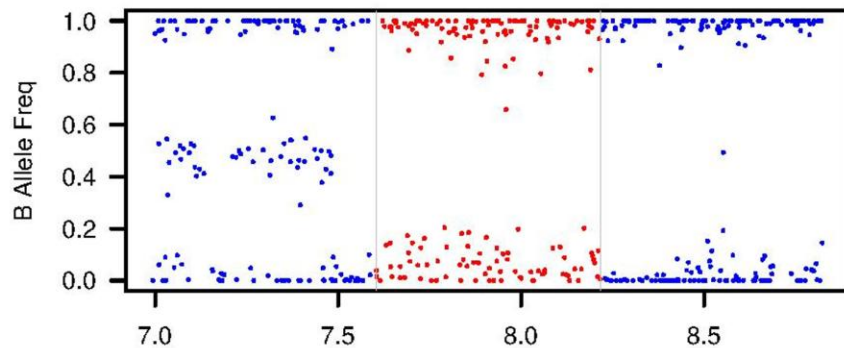
- Real Data Performance Visualization and Evaluation

CN=1, conf=579.641

CanineHD.5508694385549012621872_K9HD_D07_CAI6745A43_cdi



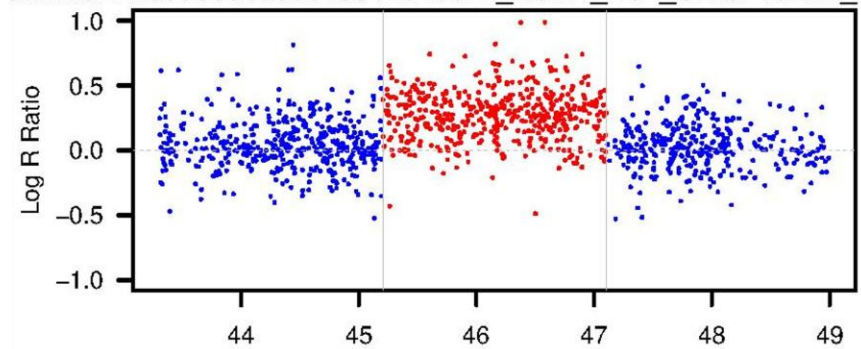
Chr31 Position (Mb)



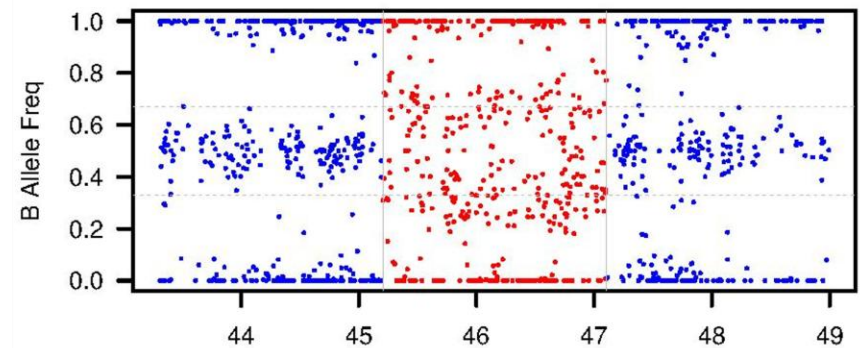
Chr31 Position (Mb)

CN=3, conf=731.338

CanineHD.5508694385549012621872_K9HD_B02_CAI6745A14_cdi



Chr6 Position (Mb)



Chr6 Position (Mb)

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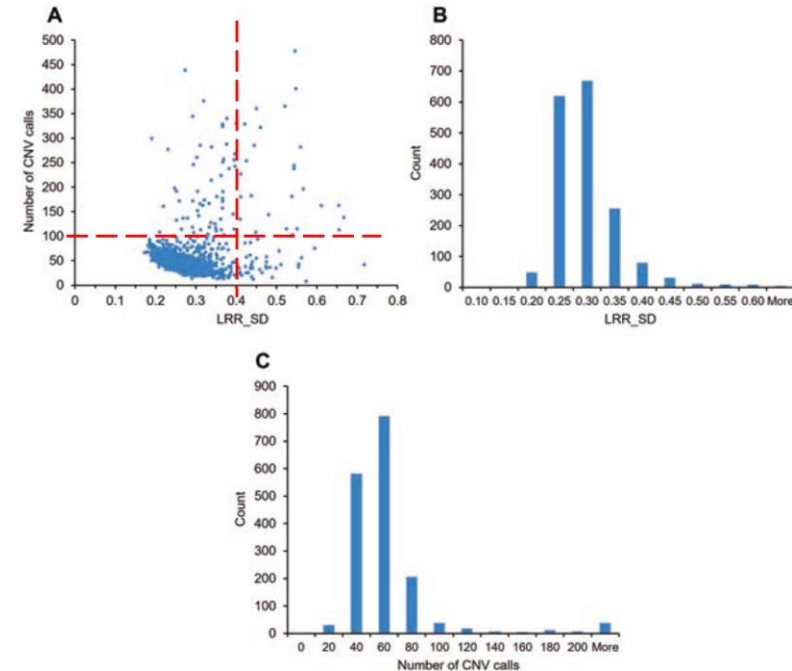
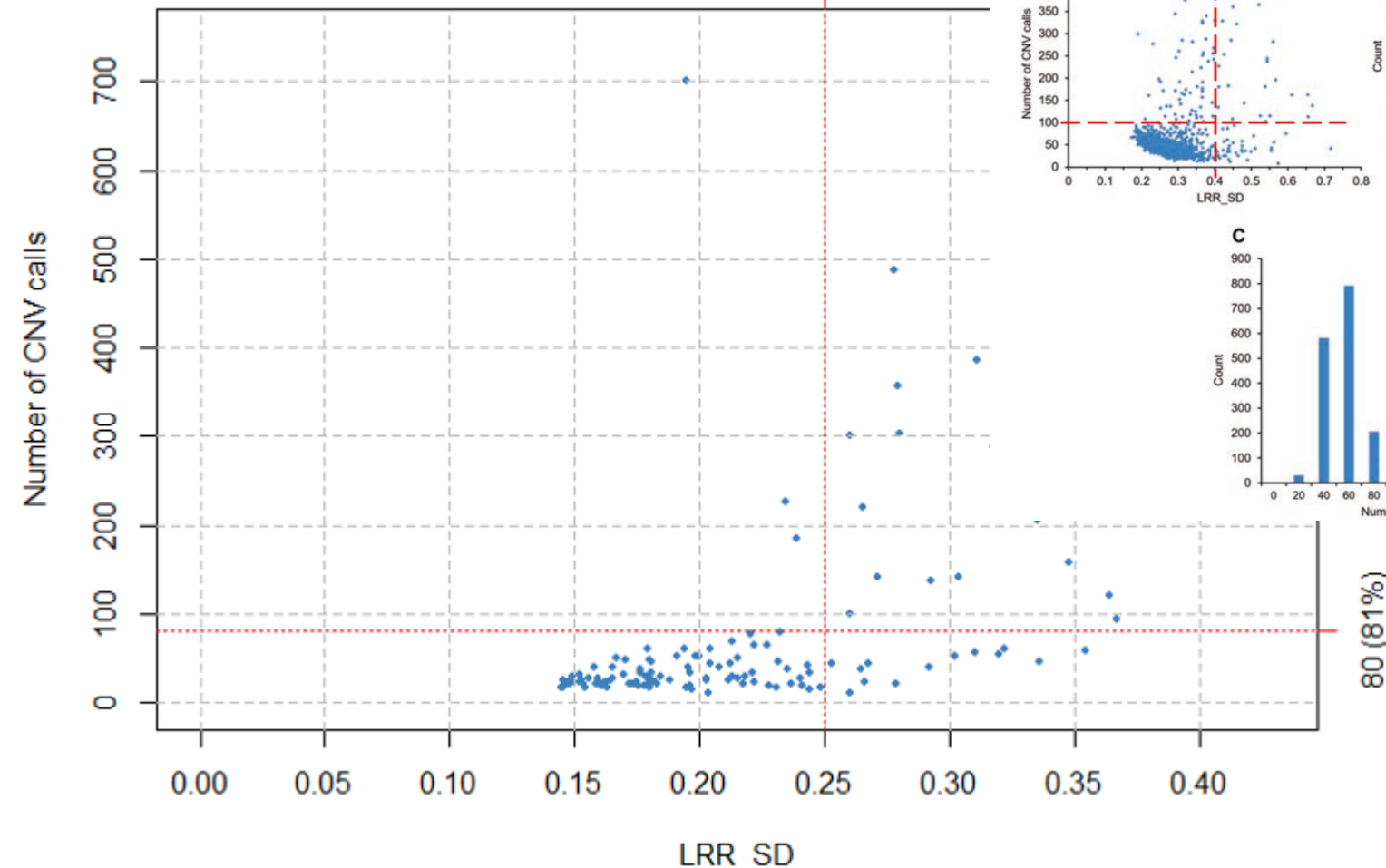
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CNV Analysis and PennCNV



- CNV Calls Filtering

0.25 (73%)

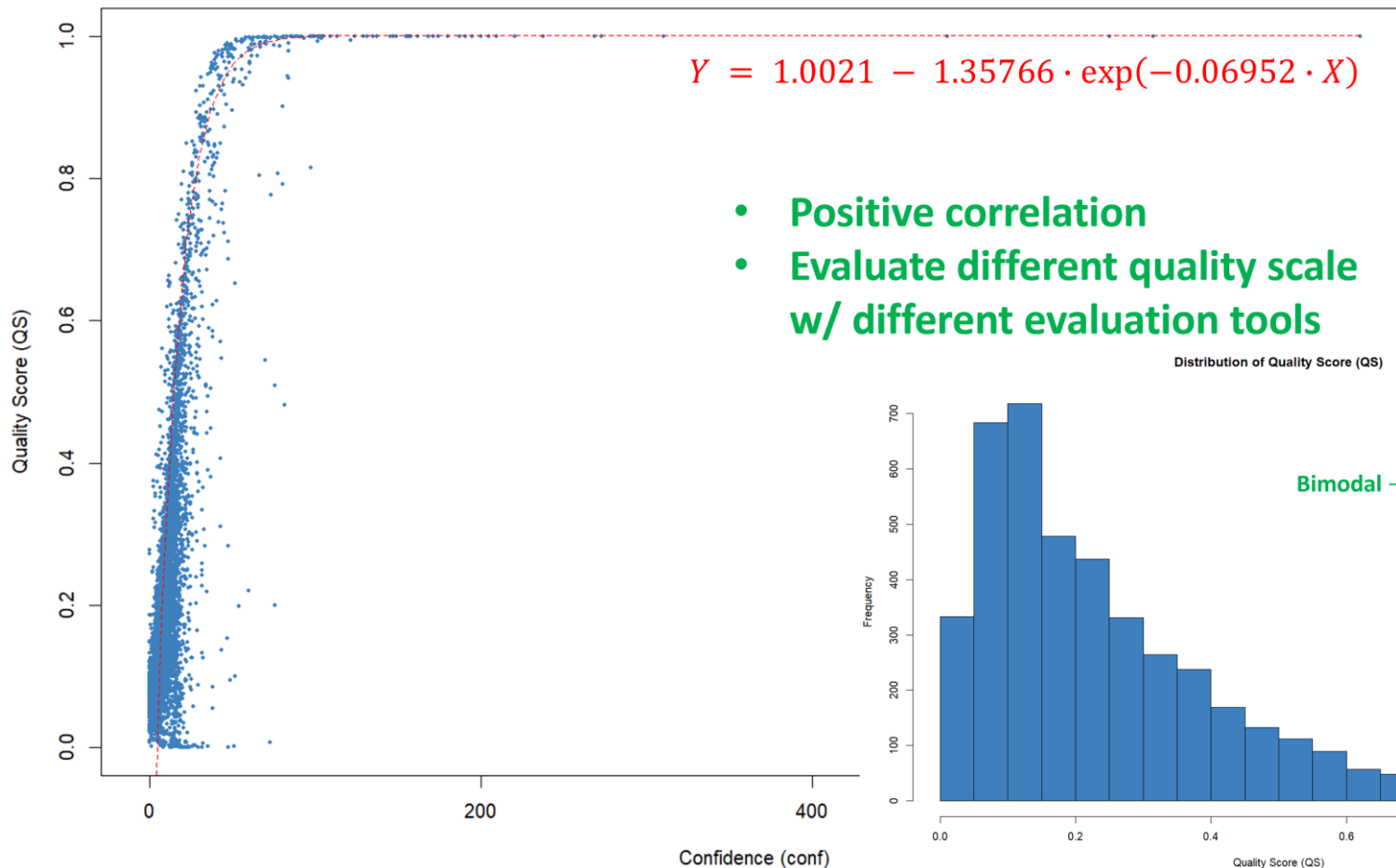


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- CNV Calls Filtering

Quality Score vs. Confidence



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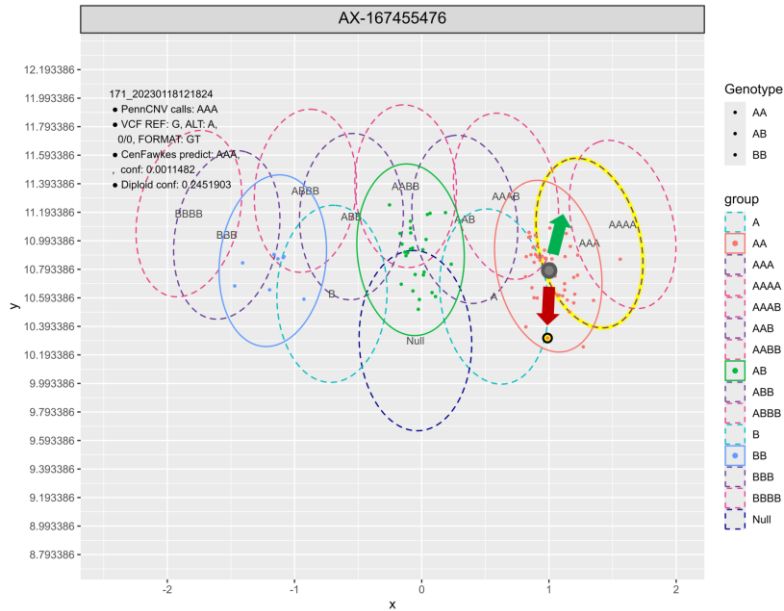
- Imputation Methods for Approximation

- $W \equiv$ A allele of some genotype cluster g , $V \equiv$ B allele of some genotype cluster g .
 $g = \text{Null, A, AA, AB, ABB, ...}$
- $T \equiv \log_e W$, $S \equiv \log_e V$
- $W \sim \text{logNormal}(\mu_t, \sigma_t^2)$, $V \sim \text{logNormal}(\mu_s, \sigma_s^2)$, $\text{cov}(W, V) = \rho_{wv} \cdot \sigma_w \cdot \sigma_v$
 $\Rightarrow T \sim \text{Normal}(\mu_t, \sigma_t^2)$, $S \sim \text{Normal}(\mu_s, \sigma_s^2)$, $\text{cov}(T, S) = \rho_{ts} \cdot \sigma_t \cdot \sigma_s$
- From T, S to W, V :
 - $E[W] = \exp\left\{\mu_t + \frac{\sigma_t^2}{2}\right\}$, $E[V] = \exp\left\{\mu_s + \frac{\sigma_s^2}{2}\right\}$
 - $\text{Var}(W) = (\exp\{\sigma_t^2\} - 1) \cdot \exp\{2\mu_t + \sigma_t^2\}$, $\text{Var}(V) = (\exp\{\sigma_s^2\} - 1) \cdot \exp\{2\mu_s + \sigma_s^2\}$
 - $\text{cov}(W, V) = \exp\left\{\mu_t + \mu_s + \frac{1}{2} \cdot (\sigma_t^2 + \sigma_s^2)\right\} \cdot (\exp\{\rho_{ts} \cdot \sigma_t \cdot \sigma_s\} - 1)$
- From W, V to T, S :
 - $\mu_t = \ln(E[W]) - \frac{1}{2} \ln\left(1 + \frac{\text{Var}(W)}{E[W]^2}\right)$, $\mu_s = \ln(E[V]) - \frac{1}{2} \ln\left(1 + \frac{\text{Var}(V)}{E[V]^2}\right)$
 - $\sigma_t^2 = \ln\left(1 + \frac{\text{Var}(W)}{E[W]^2}\right)$, $\sigma_s^2 = \ln\left(1 + \frac{\text{Var}(V)}{E[V]^2}\right)$
 - $\text{cov}(T, S) = \ln\left\{\frac{\text{cov}(W, V)}{E[W] \cdot E[V]} + 1\right\}$
- In W vs. V domain, mean intensity stretch ratio: $\mu_{ra} = \frac{(\mu_{AAa} - \mu_{ABa})}{(\mu_{ABa} - \mu_{BBa})}$ ($\mu_{rb} = \frac{(\mu_{BBb} - \mu_{ABb})}{(\mu_{ABb} - \mu_{AAb})}$),
 $\Rightarrow$ When copynumber $C > 2$, $\mu_{aC} = \mu_{a(C-1)} + \mu_{ra} \cdot (\mu_{a(C-1)} - \mu_{a(C-2)})$
- In W vs. V domain, $\sigma_{ra} = \frac{(\sigma_{AAa} - \sigma_{ABa})}{(\mu_{AAa} - \mu_{ABa})}$ ($\sigma_{rb} = \frac{(\sigma_{BBb} - \sigma_{ABb})}{(\mu_{BBb} - \mu_{ABb})}$),
 $\Rightarrow$ When copynumber $C > 2$, $\sigma_{aC} = \sigma_{a(C-1)} + \sigma_{ra} \cdot (\mu_{a(C)} - \mu_{a(C-1)})$
- Relationships b/w μ_w and σ_w , and b/w μ_v and σ_v : **linear**, parabola, exponential

CNV Analysis and PennCNV

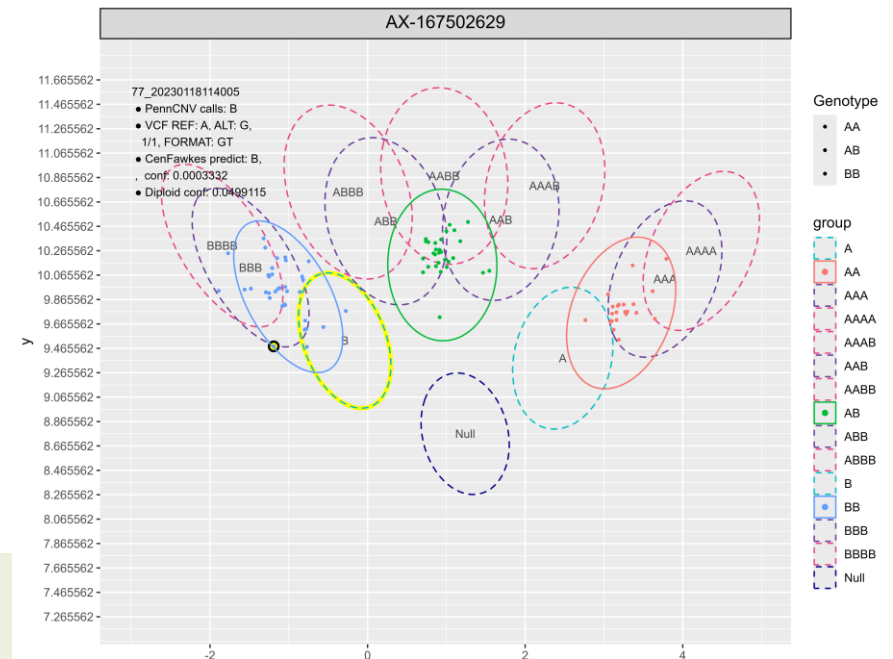


• CNV Clusters Inference



Banffdog Data

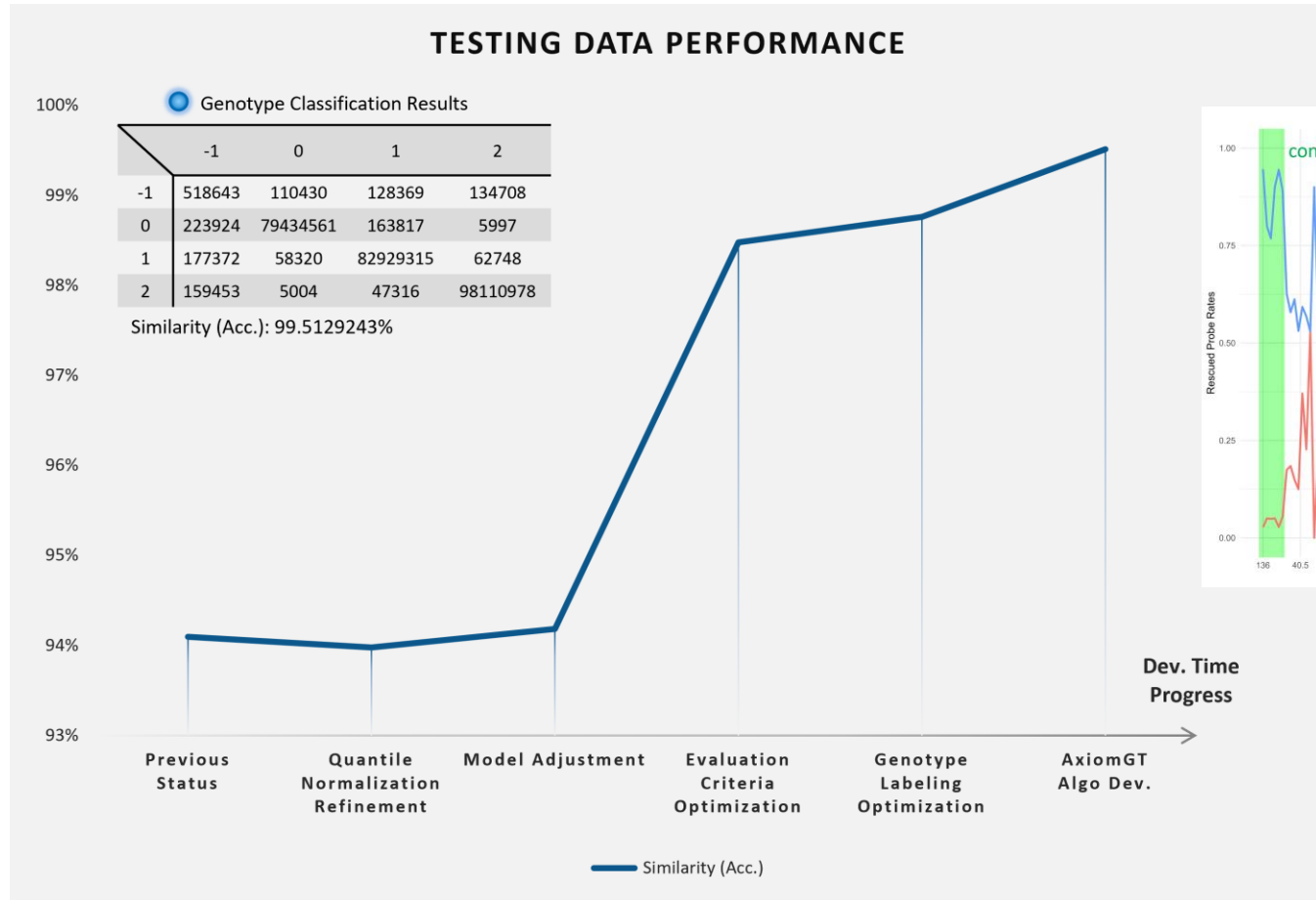
⇒ Fake (False Positive) CNV call



CNV Analysis and PennCNV



- Overall Genotyping Performance Evaluation



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CNV Analysis and PennCNV



- Development, Implementation, and Distribution

TERMINAL

PROBLEMS

OUTPUT

```
1 [|||||] 27.35 0 [|||||] 18.65 17 [|||||] 100.00 25 [|||||] 100.00
2 [|||||] 100.00 10 [|||||] 100.00 18 [|||||] 100.00 26 [|||||] 100.00
3 [|||||] 100.00 11 [|||||] 1.35 19 [|||||] 100.00 27 [|||||] 100.00
4 [|||||] 100.00 12 [|||||] 100.00 20 [|||||] 100.00 28 [|||||] 100.00
5 [|||||] 100.00 13 [|||||] 100.00 21 [|||||] 100.00 29 [|||||] 100.00
6 [|||||] 100.00 14 [|||||] 6.65 22 [|||||] 100.00 30 [|||||] 100.00
7 [|||||] 100.00 15 [|||||] 100.00 23 [|||||] 100.00 31 [|||||] 100.00
8 [|||||] 0.75 16 [|||||] 0.75 24 [|||||] 100.00 32 [|||||] 100.00
9 [|||||] 57.56/1000 119/26.00 Load average: 16.92 5.60 2.14
Tasks: 62, 320 thr, 26 running
Uptime: 2 days, 14:55:26
```

CPT

Apart from PennCNV, Centrillion Technology also provides programs to infer SNP compositions of CNVs. Different from using LRR, BAF, and HMM model to predict CNV in PennCNV, Centrillion Technology utilizes MvA intensity data and GMM model to predict CNV clusters for each candidate probeset directly without any further signal space transformation. The program adapts the Fawkes algorithm in the raw intensity (allele-summarized) domain to the MvA-transformed domain. For more information about the usage of the program and original algorithms in the following script, please refer to CPT/include/CPT/application/cnv_inference/R/CenFawkesOnMvA.R helper document, and documents of Fawkes algorithm respectively. All the input files at this step can be generated or derived from previous steps and CPT library.

```
#!/bin/bash
mkdir /home/test/test_env/CenFawkes && cd /home/test/test_env/CenFawkes

sudo Rscript --vanilla /home/test/CPT/include/CPT/application/cnv_inference/R/CenFawkesOnMvA.R \
-C /home/test/test_env/BanffDog.6Cadjusted.snp10.Length100k.filtered_sample.goodcnv \
-p /home/test/test_env/BanffDog.pfb \
-i /home/test/test_env/mva.summary.txt \
-g /home/test/test_env/genotype.calls.txt \
-M /home/test/CPT/analysis_mdL/AxiomGT1.mdl \
-o /home/test/test_env/CenFawkes \
-n 20 \
-W
```

The above command indicates inferred CNV genotypes (chr19:46625504-49653801) of sample 165_20230118123031 is stored under "/home/test/test_env/CenFawkes", and the inferred CNV genotypes will look like the following:

Name	Chr	Position	Minus	Average	GENO	VCF
AX-167366987	19	46625504	2.17053	10.2123	AAA	0/0
AX-167439331	19	46628779	0.345515	11.6491	AAB	0/1
AX-167418746	19	46652649	-2.98257	11.3367	BBB	1/1
AX-167194203	19	46709730	2.58435	10.5129	AAA	0/0
AX-168196510	19	46729581	2.36	9.94711	AAA	0/0
...						

NOTICE: To derive better and more precise results, filtered_sample.goodcnv can be artificially further filtered from some PennCNV QC metrics and the PennCNV visualization results.

Moreover, CPT also provides a visualization tool to help users validate and visualize a probeset's diploid genotyping results, imputed genotyping clusters of different copy numbers, and the CNV call of specified sample in the MvA domain at the same time. For more information about the usage and algorithms in the following script, please refer to CPT/include/CPT/application/cnv_inference/R/CenGenotypingResVisualizer.R helper document. All the input files at this step can be generated or derived from previous steps and the CPT library.

```
#!/bin/bash
cd /home/test/test_env/CenFawkes

sudo Rscript --vanilla /home/test/CPT/include/CPT/application/cnv_inference/R/CenGenotypingResVisualizer.R \
-f samples \
```